DATE :

Aplikasi Integral

A. Latihan 1

1. $\int_{-1}^{2} (3x^2 - 2x + 3)\, dx$

2. $\int_{0}^{3} (x^3 - 6x)\, dx$

3. $\int_{1}^{2} (x+1)^2\, dx$

4. $\int_{1}^{4} (t^2 - t - 6)\, dx$

5. $\int_{1}^{9} \frac{2t^2 + t^2\sqrt{t} - 1}{t}\, dt$

6. $\int_{1}^{8} \frac{t-1}{\sqrt[3]{t^2}}\, dt$

7. $\int_{-1}^{2} \frac{1}{(t+2)^2}\, dt$

8. $\int_{1}^{7} \frac{1}{\sqrt{2w+2}}\, dw$

Jawab

1. $= (x^3 - x^2 + 3x)\Big|_{-1}^{2}$

$= (8 - 4 + 6) - (-1 - 1 - 3)$

$= 10 - (-5)$

$= 15$

2. $\frac{1}{4}x^4 - 3x^2\Big|_{0}^{3}$

$\left(\frac{1}{4} \cdot 81 - 27\right) - 0$

$-\frac{27}{4}$ //

3. $\int_{1}^{2} (x^2 + 2x + 1)$

$\left(\frac{1}{3}x^3 + x^2 + x\right)\Big|_{1}^{2}$

$\left(\frac{8}{3} + 4 + 2\right) - \left(\frac{1}{3} + 1 + 1\right)$

$\left(\frac{8+12+6}{3}\right) - \left(\frac{1+3+3}{3}\right) = \frac{26-7}{3} = \frac{19}{3}$ //

4) $\int_{1}^{4} (t^2 - t - 6)\, dt$

$= \left(\frac{1}{3}t^3 - \frac{1}{2}t^2 - 6t\right)\Big|_{1}^{4}$

$= \left(\frac{64}{3} - 8 - 24\right) - \left(\frac{1}{3} - \frac{1}{2} - 6\right)$

$= -\frac{32}{3} + \frac{37}{6} = \frac{-64+37}{6}$

$= \frac{-27}{6} = \frac{-9}{2}$ //

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B

1. $\int_{\pi/3}^{\pi/2} 3\sin(2\theta)\, d\theta$

2. $\int_{0}^{\pi/2} \sin(2x-\pi)\, dx$

3. $\int_{0}^{\pi} t \sin t\, dt$

4. $\int_{0}^{1} t \cos \pi t\, dt$

Jawab.

1 $\left(-\frac{3}{2}\cos 2\theta\right)\Big|_{\pi/3}^{\pi/2} = \frac{3}{2} - \left(-\frac{3}{2}\cdot\left(\frac{-1}{2}\right)\right) = \frac{3}{2} - \frac{3}{4} = \frac{3}{4}$ //

2. misal $u = 2x-\pi \rightarrow \frac{du}{dx} = 2 \rightarrow dx = \frac{1}{2}du$

$\int_{0}^{\pi/2} \sin u \cdot \frac{1}{2} du = \int_{0}^{\pi/2} \frac{1}{2}\sin u\, du = -\frac{1}{2}\cos u\Big|_{0}^{\pi/2}$

$= -\frac{1}{2}\cos(2x-\pi)\Big|_{0}^{\pi/2}$

$= \left(-\frac{1}{2}\cdot 1\right) - \left(-\frac{1}{2}(-1)\right) = -\frac{1}{2} - \frac{1}{2} = -1$ //

3. $\int_{0}^{\pi} t \sin t\, dt$

$\int d(uv) = du\, v + u\, dv$

$\int u\, dv = u\, v - \int v\, du$

misal $u = t \rightarrow \frac{du}{dt} = 1 \rightarrow du = dt$

$dv = \sin t\, dt \rightarrow v = -\cos t$

$\int_{0}^{\pi} t\sin t\, dt = t(-\cos t) - \int(-\cos t)\, dt$

$= (-t\cos t + \sin t)\Big|_{0}^{\pi}$

$= (\pi + 0) - (0 + 0)$

$= \pi$ //

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C.

1. $\int \cos t \, dt$
2. $\int \sin t \, dt$
3. $\int \cos 2t \, dt$
4. $\int \sin 2t \sin 3t \, dt$
5. $\int \sin 2t \cos 3t \, dt$
6. $\int \cos 2t \cos 3t \, dt$
7. $\int \cos^2 t \, dt$
8. $\int \sin^2 t \, dt$

Jawab

1. $\int \cos t \, dt = \sin t + C$

- $0 < t < 2\pi$

  $\sin t \Big|_0^{2\pi} = 0 - 0 = 0$ //

- $-\pi < t < \pi$

  $\sin t \Big|_{-\pi}^{\pi} = 0 - 0 = 0$ //

2. $\int \sin t \, dt = -\cos t + C$

- $0 < t < 2\pi$

  $-\cos t \Big|_0^{2\pi} = -1 + 1 = 0$ //

- $-\pi < t < \pi$

  $-\cos t \Big|_{-\pi}^{\pi} = 1 - 1 = 0$ //

3. $\int \cos 2t \, dt = \frac{1}{2} \sin 2t + C$

- $0 < t < 2\pi$

  $\frac{1}{2} \sin 2t \Big|_0^{2\pi} = 0 - 0 = 0$ //

- $-\pi < t < \pi$

  $\frac{1}{2} \sin 2t \Big|_{-\pi}^{\pi} = 0 - 0 = 0$ //

4. $\int \sin 2t \sin 3t = \frac{1}{2} (\cos(-t) - \cos(5t)) + C$

$= \frac{1}{2} (\cos t - \cos 5t) + C$

- $0 < t < 2\pi$

  $\frac{1}{2} (\cos t - \cos 5t) \Big|_0^{2\pi}$

  $= 0 - 0 = 0$ //

- $-\pi < t < \pi$

  $= \frac{1}{2} (\cos t - \cos 5t) \Big|_{-\pi}^{\pi}$

  $= 0 - 0 = 0$ //

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Latihan 2

1. $y = 3x^2$ antara $x = 1$ dan $x = 2$

$L = \int_1^2 3x^2 dx = x^3 \Big|_1^2 = 8 - 1 = 7$ satuan //

2. $y = 2x^3$ antara $x = 0$ dan $x = 2$

$L = \int_0^2 2x^3 dx = \frac{1}{2} x^4 \Big|_0^2 = 8 - 0 = 8$ satuan //

3. $y = x^2 + 4$ antara $x = 0$ dan $x = 3$

$L = \int_0^3 (x^2 + 4) dx = \left(\frac{1}{3} x^3 + 4x\right) \Big|_0^3 = 21 + 0 = 21$ satuan //

4. $y = 6x - x^2$ antara 0 dan 6

$L = \int_0^6 (6x - x^2) dx = \left(3x^2 - \frac{1}{3} x^3\right) \Big|_0^6 = (108 - 72) - 0 = 36$ satuan //

5. $y = x^3 + 1$ antara $x = 2$ dan $x = 4$

$L = \int_2^4 (x^3 + 1) dx = \left(\frac{1}{4} x^4 + x\right) \Big|_2^4 = 68 - 6 = 62$ satuan //

6. $y = 4x^2 - x + 4$ antara $x = 1$ dan $x = 3$

$L = \int_1^3 (4x^2 - x + 4) dx = \left(\frac{4}{3} x^3 - \frac{1}{2} x^2 + 4x\right) \Big|_1^3$

$= \frac{87}{2} - \frac{29}{6} = 38\frac{2}{3}$ satuan //

7. $y = x^3 - 4x^2 + 4x$ antara $x = 0$ dan $x = 2$

$L = \int_0^2 (x^3 - 4x^2 + 4x) dx = \frac{1}{4} x^4 - \frac{4}{3} x^3 + 2x^2 \Big|_0^2$

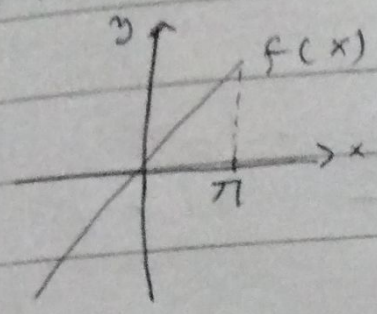
$= \frac{4}{3} - 0 = 1\frac{1}{3}$ satuan //

8. $y = \frac{10}{x^2} = 10x^{-2}$ antara $x = 1$ dan $x = 5$

$L = \int_1^5 (10x^{-2}) dx = -10x^{-1} \Big|_1^5 = -\frac{10}{x} \Big|_1^5 = -2 - (-10) = 8$ satuan

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2. $f(x) = x$



$$y_{av} = \frac{1}{\pi - 0} \int_0^{\pi} x \, dx$$

$$= \frac{1}{\pi} \left(-\frac{1}{2} x^2\right)\Big|_0^{\pi}$$

$$= \frac{1}{\pi} \left(\frac{\pi^2}{2}\right) = \frac{\pi}{2} //$$

$$y_{rms} = \sqrt{\frac{1}{\pi} \int_0^{\pi} x^2 \, dx} = \sqrt{\frac{1}{\pi} \left(\frac{1}{3} x^3\right)\Big|_0^{\pi}}$$

$$= \sqrt{\frac{1}{\pi} \left(\frac{\pi^3}{3}\right)} = \frac{\pi}{\sqrt{3}} //$$

3.

$$y_{av} = \frac{1}{0,05} \int_0^{0,05} 10 e^{-200t} \, dt$$

$$= \frac{1}{0,05} \left(10 \cdot \frac{1}{-200} e^{-200t}\right)\Big|_0^{0,05}$$

$$= \frac{1}{0,05} \left(-\frac{1}{20} e^{-200t}\right)\Big|_0^{0,05}$$

$$= \frac{1}{0,05} \left(\frac{-1}{20e^{10}} + \frac{1}{20}\right)$$

$$= -e^{-10} + 1 //$$

$$y_{rms} = \sqrt{\frac{1}{0,05} \int_0^{0,05} (10e^{-200t})^2 \, dt}$$

$$= \sqrt{\frac{1}{0,05} \int_0^{0,05} 100 e^{-400t} \, dt}$$

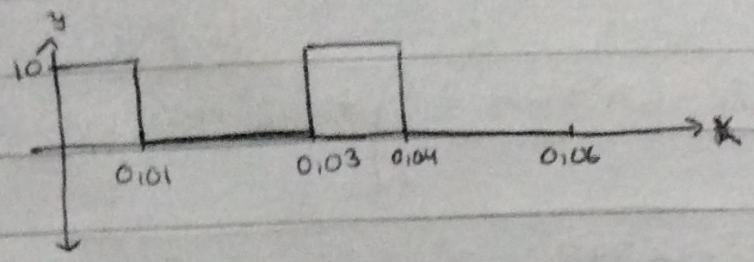
$$= \sqrt{\frac{1}{0,05} \left(100 \cdot \frac{1}{-400} e^{-400t}\right)\Big|_0^{0,05}}$$

$$= \sqrt{\frac{1}{0,05} \left(-\frac{1}{4} e^{-400t}\right)\Big|_0^{0,05}} = \sqrt{\frac{1}{0,05} \left(\frac{-1}{4e^{20}} + \frac{1}{4}\right)} = \sqrt{\frac{-5}{e^{20}} + 5} //$$

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B.

9. $f(x)\begin{cases} 10 & \text{bila } 0<x<0{,}01 \\ 0 & \text{bila } 0{,}01<x<0{,}03 \end{cases}$ $\quad T = 0{,}03$



$$y_{av} = \frac{1}{T}\int_0^{0{,}03} f(x)\,dx$$

$$= \frac{1}{0{,}03}\left(\int_0^{0{,}01} 10\,dx + \int_{0{,}01}^{0{,}03} 0\,dx\right)$$

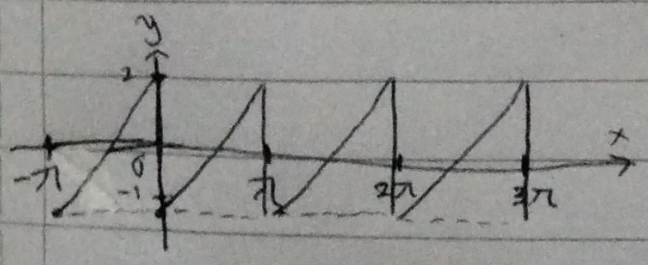
$$= \frac{1}{0{,}03}\left(10x\Big|_0^{0{,}01} + 0\right)$$

$$= \frac{1}{0{,}03}\cdot 0{,}1 = \frac{10^{-1}}{3\times10^{-2}} = \frac{1}{3}\times 10 = \frac{10}{3}\ //$$

$$y_{rms} = \sqrt{\frac{1}{0{,}03}\left(\int_0^{0{,}01} 100\,dx + \int_{0{,}01}^{0{,}03} 0\right)}$$

$$= \sqrt{\frac{1}{0{,}03}\cdot 1} = \sqrt{\frac{100}{3}} = \frac{10}{\sqrt{3}} = \frac{10\sqrt{3}}{3}\ //$$

10. $f(x)\begin{cases} x+2 & \text{bila } -\pi<x<0 \\ -x+2 & \text{bila } 0<x<\pi \end{cases}$ $\quad T = 2\pi$



$$y_{av} = \frac{1}{2\pi}\left(\int_{-\pi}^{0} x+2\,dx + \int_0^{\pi} -x+2\,dx\right)$$

$$= \frac{1}{2\pi}\left(\left[\frac{1}{2}x^2+2x\right]_{-\pi}^{0} + \left[-\frac{1}{2}x^2+2x\right]_0^{\pi}\right)$$

$$= \frac{1}{2\pi}\left(\frac{\pi^2}{2}-2\pi+\left(-\frac{\pi^2}{2}+2\pi\right)\right)$$

$$= 0$$

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$$y_{rms} = \sqrt{\frac{1}{2\pi}\left(\int_{-\pi}^{0}(x+2)^2\,dx + \int_{0}^{\pi}(-x+2)^2\,dx\right)}$$

$$= \sqrt{\frac{1}{2\pi}\left(\left[\frac{1}{3}(x+2)^3\right]_{-\pi}^{0} + \left[\frac{-1}{3}(-x+2)^3\right]_{0}^{\pi}\right)}$$

$$= \sqrt{\frac{1}{2\pi}\left(\left(\frac{8}{3} - \left(\frac{-\pi^3-6\pi^2+12\pi-8}{3}\right)\right) + \left(\frac{\pi^3-6\pi^2+12\pi-8}{3} + \frac{8}{3}\right)\right)}$$

$$= \sqrt{\frac{1}{2\pi}\left(\frac{8+\pi^3+6\pi^2-12\pi+8}{3} + \frac{\pi^3-6\pi^2+12\pi-8}{3} + \frac{8}{3}\right)}$$

$$= \sqrt{\frac{1}{2\pi}\left(\frac{16}{3} + \frac{2\pi^3}{3}\right)}$$

$$= \sqrt{\frac{2\pi^3+16}{6\pi}}$$